

Tutorial Sheet 3

Anant Kumar

October 6, 2025

1. Answer the following:

- (a) Analog signal $x_a(t) = 4 + 2 \cos(150\pi t + \pi/3) + 4 \sin(350\pi t)$ is sampled at a rate of 200 samples/sec to obtain the DT signal $x(n)$. Determine $x(n)$ and DTFT $Y(e^{j\omega})$.

Solution

$$\begin{aligned} x(n) &= x_a(nT) \\ &= 4 + 2 \cos(150\pi n/200 + \pi/3) + 4 \sin(350\pi n/200) \\ &= 4 + 2 \cos(3\pi n/4 + \pi/3) + 4 \sin(7\pi n/4) \end{aligned}$$

DTFT is given by:

$$\begin{aligned} Y(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} \\ &= 4 \sum_{n=-\infty}^{\infty} e^{-j\omega n} + 2 \sum_{n=-\infty}^{\infty} \cos(3\pi n/4 + \pi/3)e^{-j\omega n} + 4 \sum_{n=-\infty}^{\infty} \sin(7\pi n/4)e^{-j\omega n} \\ &= 4 \cdot 2\pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k) + 2 \cdot \pi \left[\frac{1}{2} e^{j\pi/3} \sum_{k=-\infty}^{\infty} \delta(\omega - 3\pi/4 - 2\pi k) + \frac{1}{2} e^{-j\pi/3} \sum_{k=-\infty}^{\infty} \delta(\omega - 3\pi/4 + 2\pi k) \right. \\ &\quad \left. + 4 \cdot \pi \left[\frac{1}{2j} \sum_{k=-\infty}^{\infty} \delta(\omega - 7\pi/4 - 2\pi k) - \frac{1}{2j} \sum_{k=-\infty}^{\infty} \delta(\omega + 7\pi/4 - 2\pi k) \right] \right] \\ &= 8\pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k) + \pi e^{j\pi/3} \sum_{k=-\infty}^{\infty} \delta(\omega - 3\pi/4 - 2\pi k) + \pi e^{-j\pi/3} \sum_{k=-\infty}^{\infty} \delta(\omega + 3\pi/4 - 2\pi k) \\ &\quad + 2\pi \left[\frac{1}{j} \sum_{k=-\infty}^{\infty} \delta(\omega - 7\pi/4 - 2\pi k) - \frac{1}{j} \sum_{k=-\infty}^{\infty} \delta(\omega + 7\pi/4 - 2\pi k) \right] \end{aligned}$$